Numerical optimization for self-tuning electrohydraulic control systems

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Abstract

In this paper we present an approach to self tuning control system for hydraulic actuators with flexible mechanical loads. Explicit criteria optimization using the COMPLEX optimization algorithm is used to adjust the regulator parameters towards their optimal values.

The regulator structure is obtained by considering the problem in a linear state space formulation and by computing a regulator based on state feedback and state estimation. Results from experiments carried out on a lorry crane are presented.

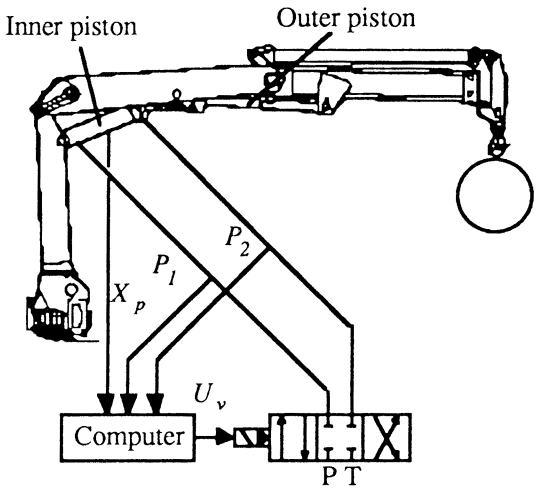


Figure 1: Computer controlled crane..

1 Introduction

Systems including fluid power components are interesting applications of adaptive control since they are nonlinear and the dynamics change when the operating point of the system changes. It is therefore difficult to design regulators that work over the whole range of operating points.

Some applications of fluid power systems involve flexible mechanical loads, for example in mobile hydraulics, and this makes the problem more complex. Studies of this kind of systems have been presented in Krus and Gunnarsson [1990] where adaptive control was used for hydraulic actuators with flexible loads.

Here, a numerical optimization technique is used to optimize the regulator parameters explicitly. This means that there is no need to establish a model in order to calculate the regulator parameters. The technique is thus very straightforward and can be used for selftuning regulators. Numerical optimization cannot, however, be used for adaptive control unless the load cycle is repetitive.

Numerical optimization of control systems and control strategies for hydraulic systems using dynamic simulation has been described by several authors, ie Sankar and Blach (1980) who used it for the optimization of control parameters in a hydrostatic vehicle drive, Rydberg (1983) who also used it for the optimization of hydrostatic drives.

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Roth (1983) used it for optimizing control parameters for position control of hydraulic servo systems using the DSH package developed at RWTH in Aachen. In Krus et al. (1991) it was shown how optimization could be used in system design to make an overall optimization which included most parameters of the system.

The work has been carried out using computer simulations and experiments on a real lorry crane, which is depicted in Fig. 1.

Due to the flexibility in the system the position of a load at the tip of the crane will show an oscillatory behavior, and a main goal will hence be to design a regulator that gives a well damped system.

2 Regulator design

The crane shown in Fig. 1 represents a dynamic system of high complexity, which would be very difficult to model in detail. However, the simplified model given by Fig. 2 represents a fair description of the dynamics of the crane.

Using this model it has been assumed that the flexibility in the system is concentrated to the arm of the crane.

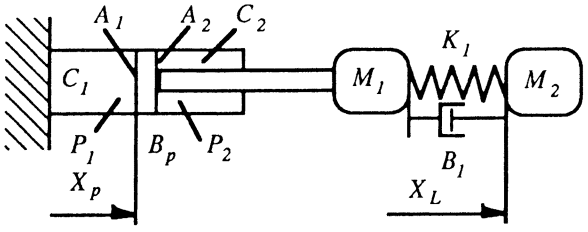


Figure 2: Simplified model of the crane.

The dynamic properties of the system are dominated by two poorly damped complex poles, and they give the system a dominating resonance frequency, ω_0. A main goal for the regulator design will hence be to reduce the oscillatory behavior of the system.

The regulator design starts from the state space description of the system, see Krus, Gunnarsson (1990). A systematic feedback design method for a system given on state space form is state feedback from estimated states, when the feedback gains are obtained using optimization. For a thorough treatment of such, so called, LQG regulators we refer to Friedland (1986).

It is often of interest to describe the feedback using transfer functions, and straightforward calculations give that the regulator based on state feedback and state estimation can be written

$$U(s) = F_r(s)X_r(s) - F_y(s)Y(s) \qquad (1)$$

$F_y(s)$ is a row vector with two transfer functions as elements since we can measure two output signals which means that

$$F_y(s) = (F_{X_p}(s) \;\; F_P(s)) \qquad (2)$$

where $F_{X_p}(s)$ and $F_P(s)$ denote the transfer functions from piston position x_p to input signal and effective load pressure P_{Le} to input signal respectively. A graphical description of the regulator structure is given in Fig. 3.

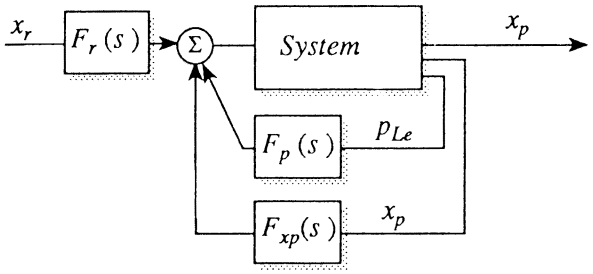


Figure 3: Regulator structure

The design of the LQG regulator described above requires complete knowledge of the system dynamics. It is, however, not realistic to assume such knowledge in our application, and we have instead used the LQG regulator to get suggestions about a suitable structure of the regulator. This has been done by assuming reasonably realistic values of the parameters in the model and then computing gain vectors that gives a well damped closed loop system. The optimization criteria has

been to minimize J defined as

$$J = \int_{t_1}^{t_2} (x_{p1r} - x_p 1)^2 + p_{Le}^2 dt \qquad (3)$$

Using these gain vectors we have then computed the transfer functions of the regulator. Two observations can then be made concerning the properties of the LQG regulator. The most important observation was that the transfer function $F_P(s)$ had band pass filter properties around ω_0. The band pass character is a desirable property since the rejection of low frequency signals cope with the DC level in the pressure measurement, while the rejection of high frequencies reduce the influence of high frequency dynamics.

Based on these observations we propose a low order regulator having the following structure.

$$U(s) = F_r(s)X_r(s) - F_{X_p}(s)X_p(s) - F_P(s)P_{Le}(s) \qquad (4)$$

where $F_{X_p}(s)$ and $F_r(s)$ are simply

$$F_{X_p}(s) = F_r(s) = K_e \qquad (5)$$

$F_p(s)$ on the other hand is a band pass filter with zero static gain

$$F_p(s) = \frac{K_p s(\frac{1}{\omega_{p1}} + \frac{1}{\omega_{p2}})}{(s/\omega_{p1} + 1)(s/\omega_{p2} + 1)} \qquad (6)$$

In this way the error introduced by the feedback of a steady state pressure is removed as well as the influence of high frequency dynamics.

3 The Complex Algorithm

indexComplex algorithm The optimization of a control system can be described as a multivariable constrained optimization problem. The problem is to maximize the function

$$F(X_1, X_2, ..., X_N)$$

subjected to the constraints

$$G_i \leq X_i \leq H_i$$

where $i = 1, 2, ..., M$. The implicit variables $X_{N+1}, ..., X_M$ are dependent functions of $X_1, ...X_N$. The constraints G_i and H_i are either constants or functions of $X_1, ...X_N$. One method that is very suitable for this kind of problem is the Complex method of Box (1965). In the implementation used here, an initial complex of k points is generated. The variables at each point are generated using random numbers.

$$x_{ij} = G_i + r_{ij}(H_i - G_i) \qquad (7)$$

Here j is an index that indicates a point in the Complex and i an index that indicates a variable. r_{ij} is a random number in the interval [0,1]. If the implicit constraints are not fulfilled, a new point is generated until the implicit constraints are fulfilled. The number of points k in the complex must be $k \geq N + 1$ where N is the number of independent variables.

The object function is evaluated at each point. The point having the lowest value is replaced by a point reflected in the centroid of the remaining points by a factor α.

$$x_{ij}(new) = \alpha(x_{ic} - x_{ij}(old)) + x_{ic} \qquad (8)$$

The centroid is calculated as

$$x_{ic} = \frac{1}{k-1}[\sum_{j=1}^{k} x_{ij} - x_{ij}(old)] \qquad (9)$$

Box (1965) recommends $\alpha = 1.3$. If a point repeats as the lowest value on consecutive trials, it is moved half of the distance to the centroid of the remaining points.

There also exist a variety of modified versions of the Complex algorithm. Notable are Manetsch (1990). Guin (1968) pointed out some problems associated with the original complex method and the way used here to overcome these are described in Krus et al (1992).

The convergence properties of the COMPLEX-method can not be calculated exactly since they depend on the problem to a great deal. Some observations can, however, be made. The most straightforward case to analyse is when the optimum is located close to the centroid of the complex, and the surface is almost part of a sphere. Each new point that is replaced then has to be

calculated two times. The first time when it is reflected through the centroid it will obtain a worse value than before, since it will be moved further away from the centroid/optimum, and not until it has be moved halfway towards the centroid will it obtain a better value. Let the average distance between the points in the complex be Δx. After we have moved all the points in the centroid once towards the centroid (after $2k$ evaluations) it will be reduced as:

$$\Delta x_{n+k} = \Delta x_n(\frac{\alpha}{2}) \tag{10}$$

where k is the number of points in the complex The reduction after only one iteration is then in average:

$$\Delta x_{n+1} = \Delta x_n(\frac{\alpha}{2})^{\frac{1}{2k}} \tag{11}$$

The reduction after i evaluations of the object function will thus be:

$$\Delta x_i = \Delta x_0(\frac{\alpha}{2})^{\frac{i}{2k}} \tag{12}$$

Given the reduction ratio $\Delta x_i/\Delta x_0$ the number of evaluations can be calculated as:

$$i = 2k \ln \frac{\Delta x_i}{\Delta x_0} / \ln \frac{\alpha}{2} \tag{13}$$

From this simple relationship we can see that the number of evaluations to reduce the complex a certain amount, is directly proportional to the number of points k in the complex. In reality the object function is rarely as simple as assumed but equation (13) sets a lower bound to the number of evaluations necessary.

4 Experimental Results

Experiments have been carried out on a real lorry crane, controlled by electrohydraulic valves, see Fig. 1. The crane was equipped with transducers for measurement of pressure and piston position. A Macintosh II fx computer was used for measurement and control, and the sampling rate used in the experiments was 33 [Hz]. The valves had overlap which had to be compensated for. Otherwise it would result in excessive error. Unfortunately these are not very well defined since they are temperature dependent. In order to identify the overlaps (in positive and negative direction) they were included among the optimization variables.

When using optimization in a situation with real hardware the process is complicated by the fact that the object function may be contaminated with noise and/or the parameters of the system may be time varying such as the temperature. This means that we cannot rely on getting the same value for the object function if it is evaluated with the same parameter set two times. As a consequence there may be no point that gives a better value than the worst point in the complex (that was evaluated some time ago), and the algorithm may hang up even if it tries the parameter set corresponding to the best point in the complex.

The approach used here to solve this dilemma is to introduce a penalty for old evaluations of the object function. After each iteration the object function values are decreased a certain amount.

In order to determine in what way the object function can be decreased. The behaviour of the object function has to be investigated. Near the optimum the object function is assumed to be

$$F_e = F - F_{opt} = -C|x - x_{opt}|^{\kappa} \tag{14}$$

Now let us consider the object function with the worst (minimum) value.

$$F_{e,min} = F_{min} - F_{opt} = -C|x_{min} - x_{opt}|^{\kappa} \tag{15}$$

Assuming that $x_{min} - x_{opt}$ is proportional to Δx in (15) and using (11) yields

$$F_{e,min}(n+1) = F_{e,min}(n)(\frac{\alpha}{2})^{\frac{\kappa}{2k}} \tag{16}$$

Subtracting a factor $\frac{\gamma}{2k}$ from the exponent yields.

$$F_{e,min}(n+1) = F_{e,min}(n)(\frac{\alpha}{2})^{\frac{\kappa-\gamma}{2k}} \tag{17}$$

This equation is convergent as long as

$$\gamma < \kappa \tag{18}$$

This can be rewritten as

$$\Delta F_{e,min}(n+1) = F_{e,min}(n)(\frac{\alpha}{2})^{\frac{\kappa}{2k}} + \Delta F_{e,min}(n)(\frac{\alpha}{2})^{\frac{\kappa}{2k}}(\frac{\alpha}{2}^{\frac{-\gamma}{2k}} - 1) \tag{19}$$

Since it is always true that

$$\Delta F_n < -F_{e,min,n} \tag{20}$$

we can after each evaluation subtract a factor F_c to each function value, as in equation (21)

$$F_{i,n+1} = F_{i,n} - F_{c,n} \tag{21}$$

where

$$F_{c,n} = \Delta F_n (\frac{\alpha}{2})^{\frac{\kappa}{2k}} (\frac{\alpha}{2}^{\frac{-\gamma}{2k}} - 1) \tag{22}$$

Here κ is in general unknown but usually

$$\frac{\kappa}{2k} << 1$$

and the following simplified expression for equation (22) can usually be used

$$F_{c,n} = \Delta F_n (\frac{\alpha}{2}^{\frac{-\gamma}{2k}} - 1) \tag{23}$$

Convergence will still be maintained as long as $\gamma < \kappa$.

Note that ΔF is the same before and after the operation in equation (21), so ΔF_{n+1} is known before F_{n+1}. Near the optimum $\kappa = 2$. It can, however, be less in other areas so it is wise not to let γ be to large. In our example $\gamma = 0.8$.

Using this procedure some drift in the object function and the optimal parameter set can be tolerated. However, since F_x is a function of ΔF, large drift and a small tolerance on the function values (small ΔF) are incompatible. On the other hand, if there is drift in the parameters there is not so much point in finding an optimum within small tolerances since it change all the time. There is, however, one action that can be made in a situation where the algorithm seems to 'hang up', and this is to reevaluate all the points in the complex. This is not an ideal solution, however, since the small radius of the parameter sets will prevent quick adjustments to a new optimum. The 'hang up' condition normally reflects that the tolerance on the function values is to narrow.

The object function had the form:

$$F = -\int_{t_1}^{t_2} |x_{p1r} - x_p 1| dt - \int_{t_1}^{t_2} |p_f| dt \tag{24}$$

where p_f is highpass filtered load pressure. In this way pressure variations are minimized together with the position error.

Since some of the independent variables of the optimization can vary over several orders of magnitudes, it can be preferable to let the Complex algorithm work with the logarithms of some the parameters instead of the parameters themselves. The variables used for the optimization and there allowed intervals are listed bellow. Here D_{b+} and

$$\begin{array}{lcll} K_x & = & \log k_{Xp} & \in [0,2] \\ K_p & = & \log k_p & \in [-8.5,-6.5] \\ \Omega_{p1} & = & \log \omega_{p1} & \in [-0.5,2.5] \\ \Omega_{p2} & = & \log \omega_{p2} & \in [-0.5,2.5] \\ D_{b+} & \in & [0.2,0.6] & \\ D_{b-} & \in & [0.4,0.8] & \end{array}$$

D_{b-} are deadband for positive and negative signals respectively.

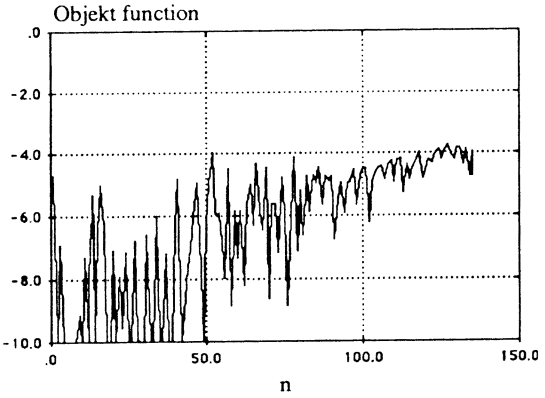


Figure 4: Object function

The development of the object function as a function of the number of simulation are shown in Fig 4. The object function is shown for all evaluations (simulations) also for the unsuccessful iterations. This explains the spiky behaviour. The first 38 evaluations represent the initial random search for an initial complex in the allowed region.

Fig 5 shows the variations in the gains K_x and K_p. Fig 6 shows the adjustment of the deadband com-

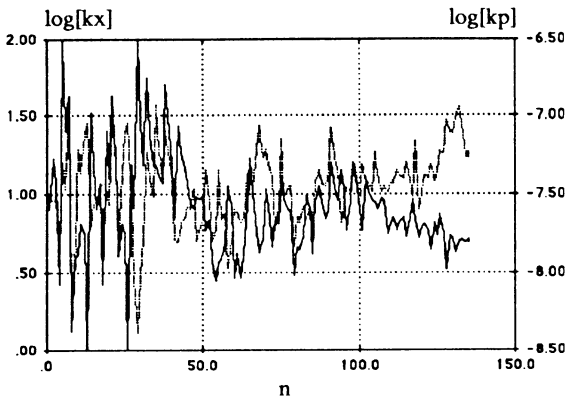


Figure 5: Adjustment of gains

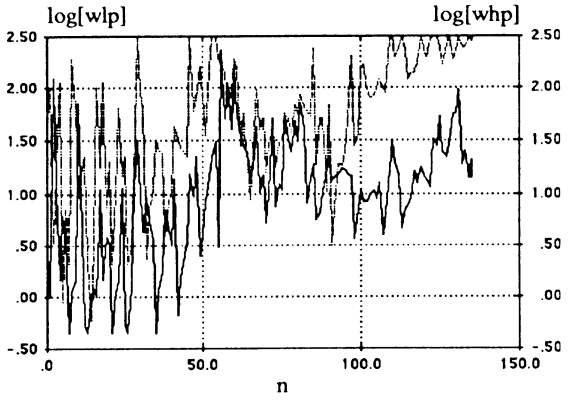


Figure 7: Adjustment of filter break frequencies for the pressure feedback

pensations. Note that variations in the deadbands are tracked between run 100 to run 150. The deadbands increase with time due to increased temperature which changes the resistance in the valve actuator spools. Fig 7 shows the variations in the filter break frequencies ω_{p1} and ω_{p2}. Note that one of the frequencies hit the upper bound. This is probably a consequence of the fact that the valve itself act as a low pass filter. Consequently the system is rather insensitive to the upper break frequency as long as it is higher than the bandwidth of the valve.

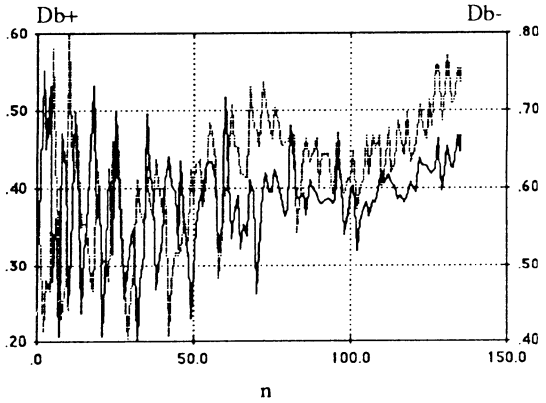


Figure 6: Adjustment of deadband compensation

The response of the optimal system is shown in Fig 8.The response of the same system but without pressure feedback is shown in Fig 9.

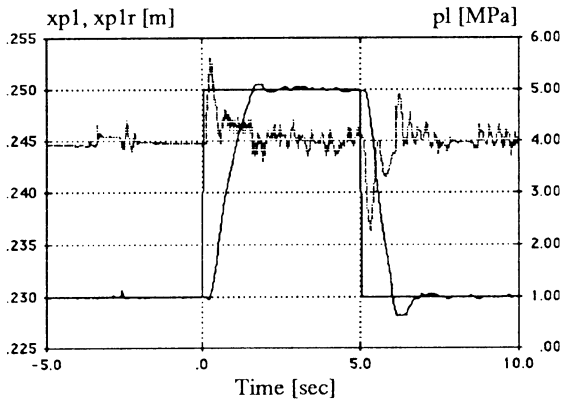


Figure 8: Optimal response

5 Discussion

Normally this kind of optimization technique is something that may be used in the start up phase, since a precisely repetitive input signal is required and this is usually not available in normal operating conditions. There are, however, some exam-

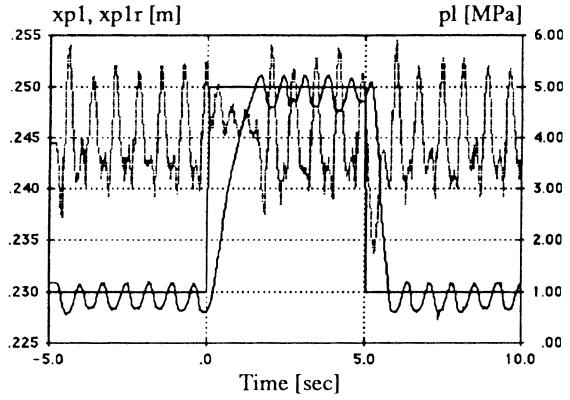


Figure 9: Step response without pressure feedback

ples, especially in manufacturing, where repetitive processes are common. In these situations this kind of optimization could be used on line. In this situation there is no need to stop the optimization. On the other hand it would be preferable to keep the parameter sets somewhat separated in order to be able to respond to changes in the process. This can be achieved by using a constant value on F_c.

The use of a progressive reduction of old object function values can also be considered for optimization where the object function is evaluated using simulation. Preliminary investigations suggests that there may be some advantages in cases where the object function is ill conditioned, ie where there are very different magnitude of gradients in different directions. It should also be pointed out that the convergence rate of the parameter sets is almost the same with and without the progressive reduction of the object function, as long as the convergence criterion in equation (18) is fulfilled.

6 Conclusions

In this paper it has been demonstrated how self-tuning using the COMPLEX-algorithm can be used for optimizing the (nonlinear) control system of an hydraulic crane.

LQG regulator synthesis was adopted establishing the regulator structure. Even though it is not practical to use the LQG regulator structure directly, since it has very high order and consequential a lot of parameters, it provided us with some useful hints on how the regulator should be designed, such as the band-pass filtering of the pressure signal.

Using the COMPLEX-algorithm it is possible to determine an optimal parameter set even though the system to be regulated is non-linear. It is also straightforward to optimize parameters related to compensation of nonlinearities such as for dead band compensation.

A disadvantage with this kind of optimization is that the system has to be run an identical load cycle many times in order to converge. This means that it can not be used for adaptive control, unless the normal operating conditions are repetitive load cycles. However, if the training cycle covers all points of operation for the machinery the resulting regulator will be robust in the sense that it is not altogether bad for any part of the work space.

An advantage is that there are very few parameters to set before the optimization (as opposed to most implementations of adaptive control) and is reliable in its convergence. It can thus serve as a test to see what the best result is, that can be achieved by a certain regulator structure in a real situation. We could also confirm that the regulator structure that established using the LQG-design made sense. None of the involved parameters had extreme values after the optimization.

7 Acknowledgements

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